

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We quantify how physical scale and population size change the emergence of complex behavior in open-ended ecological environments. Agents in our setting receive no explicit rewards or learning objectives; instead, policies evolve through reproduction, mutation, and selection while agents continuously reshape both the resource landscape and the surrounding population. Rather than optimizing a single policy, we compare emergent behavioral patterns across methods, sensors, terrains, world sizes, and seeds to measure when ecological pressure reliably produces new strategies. Using modern hardware and a new multi-agent simulator, we scale beyond prior ecological simulations to worlds with over 60,000 agents, each carrying its own evolved neural network policy. Across these runs, the empirical comparisons reveal several repeatable behaviors: long-range resource extraction, vision-based foraging, and predation emerge under competitive and survival pressures. The quantitative patterns are strongest when we compare sensor ablations and environmental scales directly: some behaviors appear only at sufficiently large population and map sizes, and larger settings consistently increase the stability and cross-seed reliability of the observed behaviors. Although evolutionary computation has a long history, these scaling results show that modern hardware can make ecology a practical empirical instrument for machine learning, enabling controlled comparisons over many agents, terrains, metrics, and evaluation conditions. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, researchers have used quantitative and comparative evidence to study how biodiversity and complex behavior emerge, spanning evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Across these disciplines, models of emergent dynamics range from theories of speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) to mathematical analyses of population genetics (Fisher 1930; Wright 1931).

Decades of mathematical modeling, laboratory experiments, and field work have produced strong evidence about the mechanisms of evolution in nature, but the same evidence becomes harder to obtain as ecosystems grow in scale. Larger and more complex ecosystems are difficult to control, measure, and replicate, and even feasible field interventions can risk damaging or displacing wild populations. These constraints make controlled, repeated comparisons across environmental scale, population size, and sensory conditions especially challenging in natural systems.

To measure how complex behavior emerges under competitive and environmental pressures, we study the open-ended evolution of neural network policies in large-scale ecological simulations. Agents have no specified objective or reward signal; instead, they collect resources to survive and reproduce under the environment dynamics, and policy variation enters only through mutations inherited from parent to child. This setup lets us compare populations of simple digital organisms across controlled changes in scale, terrain, and sensor configuration while preserving the central ecological mechanism of selection through survival and reproduction.

We conduct these comparisons in a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. The largest experiments include over 60,000 individual agents and more than 1,000,000 grid cells, making it possible to compare behavioral emergence at scales well

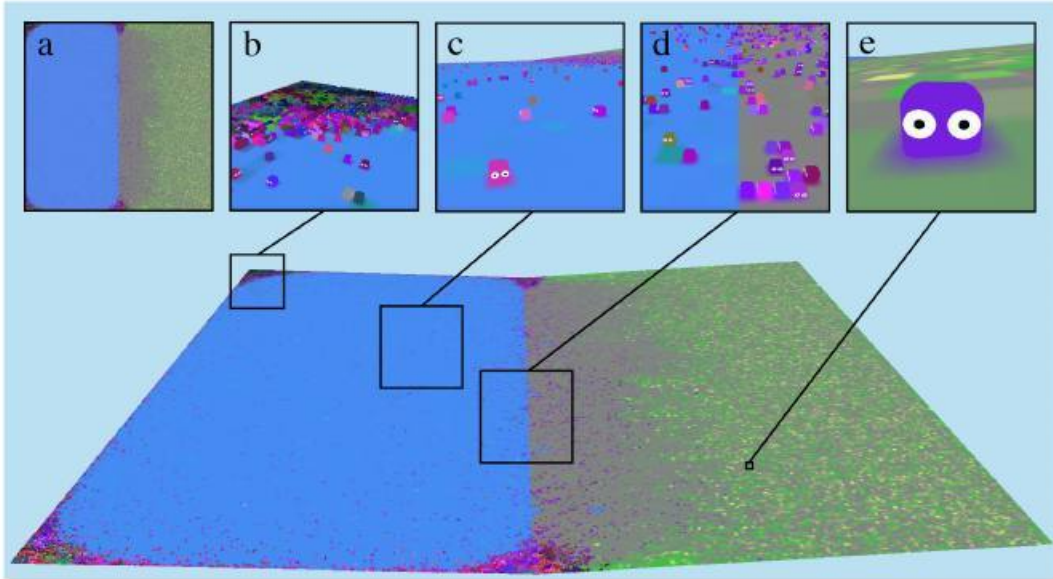


Figure 1: A 3D visualization of a 512×512 map used in our experiments. The panels make the evaluation setting explicit: inset (a) shows the full map from above, (b) shows a high-density cluster of agents in a corner, (c) shows agents in water, (d) shows agents traveling between land and water, and (e) shows an individual agent gathering resources.

beyond smaller embodied-learning testbeds. Agents must navigate to find food and other resources, and they reproduce only after collecting enough resources to create a mutated copy of themselves. Figure 1 shows a 3D visualization of the environment used for these measurements.

Our goal is both to understand ecological phenomena through simulation and to evaluate ecology as a mechanism for producing machine intelligence. Recent AI results suggest that scale can change capabilities rather than merely improve existing ones (Wei et al., 2022a), yet many embodied learning studies still use comparatively small environments and low population counts. We therefore ask what empirical patterns appear when scale is applied not to a single model, but to an entire evolving ecosystem. Analogous to capability thresholds in individual models (Wei et al., 2022b), our experiments compare whether particular strategies become more frequent, more stable, or more practically useful only when environmental size and population complexity are sufficiently large.

Across the experiments, direct comparisons show that sensor configurations and environmental scale have large and persistent effects on emergent behavior and downstream population dynamics. Compass-equipped agents learn long-range resource gathering more reliably than agents without a compass, with the margin most visible at larger map sizes; vision-equipped agents show stronger foraging and attacking patterns than otherwise comparable blind agents, again with more stable effects at larger scales. These quantitative patterns support the practical value of modern accelerators for studying how ecological evolution can produce rich behaviors under controlled, repeatable evaluation conditions.

2 RELATED WORK

Computational studies of life have long used simple rules and measurable population outcomes to investigate emergence. Conway’s Game of Life (Games, 1970) showed how local update rules can generate population-level structure, while Tierra (Ray, 1992) and Avida (Brown, modeled life as self-replicating programs that evolve through mutation. Similarly, Reynolds (1987) demonstrated that local motion rules can yield flocking and schooling. These systems establish the comparative template for our work: vary the mechanisms available to agents and measure which collective behaviors become stable.

Evolutionary methods have also produced measurable changes in the bodies and sensors of virtual agents, including locomotion morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing systems (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025). Neuroevolution has further shown that neural network architectures can be evolved directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our empirical emphasis differs from these lines of work: rather than optimizing agents toward a reward or objective, we compare what behaviors naturally arise from ecological pressure and how those behaviors change with environmental scale and complexity.

Evolution has also inspired optimization algorithms whose evidence is usually framed by performance on fixed objectives. Genetic algorithms (Holland, 1992; Jong, 1975), evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018), and novelty-oriented methods (Lehman & Stanley, 2011; Cully et al., 2015) use selection and mutation to search for high-performing or diverse solutions. These techniques have optimized neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). We also use evolutionary mechanisms to update agents, but our comparisons are not objective-function benchmarks; they measure emergent population behavior, extinction, resource use, and action statistics under controlled ecological conditions.

The reinforcement learning literature provides complementary comparisons because it often measures how task performance changes with environment complexity and agent population size. Large-scale learning environments show that environment structure can elicit broad strategies, while our experiments isolate a different evidence question: whether objective-free ecological dynamics become more reliable and behaviorally rich as map scale and population scale increase.

Simulated environments have also been central to reinforcement learning benchmarks, where quantitative comparisons are typically organized around standardized tasks. Large-scale simulators and games support single-agent and multi-agent reinforcement learning comparisons, but our contribution is complementary: we retain controlled measurement while replacing fixed task rewards with population survival, reproduction, and ecological feedback.

A closer group of studies uses biological inspiration to develop organisms within environments. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Damber (1998), and Tisue et al. (2004). Recent cellular-automata systems grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to study language and communication. Autocurricula (Leibo et al., 2019) evaluate how interacting populations generate increasingly challenging learning landscapes for one another in smaller populations (Leibo et al., 2018; Team et al., 2021). Additionally, Wang et al. (2019) and Aki et al. (2024) take the reverse approach, evolving environments specifically optimized for ideal agent learning. These works motivate our comparison of environmental conditions because they show that environment structure can materially change the behaviors agents acquire, even when the optimized object is the environment rather than the agent.

The most direct baselines for our setting are systems that simulate populations of agents interacting in complex environments. Baker et al. (2019) study Hide-and-Seek with a small number of reinforcement-learned hiders and seekers that use environmental tools, and other work trains populations for video games (Jaderberg et al., 2019), object manipulation (Petrenko et al., 2021), or capture tasks (Zheng et al., 2017; Yamada et al., 2020). However, these studies generally frame emergence under fixed tasks and objectives. Neural MMO (Suarez et al., 2019, 2023) moves closer to open-ended survival by placing agents in a massively multi-agent game and optimizing survival rather than a narrow task, but its populations remain capped at a fixed number of agents, with experiments up to 128 concurrent agents and without generational inheritance. Hamon et al. (2023) is closer still because it includes variable population sizes, genetically inspired evolution, and generational timelines, but its environment is smaller, less feature-rich, and does not directly measure scale effects. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but operates at a maximum scale of 256 agents and emphasizes tool use through programmable robots. Our experiments combine naturally varying population size, objective-free evolution, richer terrain

and resource dynamics, and substantially larger scales, allowing direct measurement of ecological phenomena that are difficult to access in these prior settings.

Finally, recent work on simulator speed and scale shows how optimized libraries can expand the empirical reach of game-like environments (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). We build on this computational direction with JAX-based acceleration, using it not only to run faster simulations but also to support controlled comparisons across grid sizes, populations, sensors, terrains, and random seeds in ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We model agent–environment interaction as an **Ecological Game (EG)**, a multiplayer formalism designed to make population-level measurements explicit. Like Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG has Markovian transition dynamics. Unlike reward-based formulations, it represents a changing population in an objective-free setting: it contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and a set of actions for each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . The key comparison with an SG is that the reward function is removed and replaced by a population function $\Gamma(s_t)$, which records which agents in I are alive and identifies their parents so that births, deaths, and lineages can be measured. As with POMDPs and POSGs, an EG can be partially observable; we call this variant a POEG. A POEG does not expose the full state to each agent, and instead uses an observation function $O(s_t, i)$ mapping a state and player to that player’s observation. Our environment is a POEG, and the experiments compare different observation functions through the sensor configurations described below.

3.2 ENVIRONMENT

The POEG used in our experiments is a large grid world in which agents must collect resources to survive and reproduce. The environment state is factored as $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m stores map-level data such as terrain and resource distributions, and s_t^a stores agent-level data such as position, health, and internal resources. This factorization supports direct measurement of both ecological quantities, such as free biomass, and population quantities, such as survival and action rates.

Terrain: Each map contains a rock layer defining the height of every grid cell, and water is initially assigned to cells whose rock value falls below a preset sea level. During an episode, water flows downhill through a discrete flow simulation based on this height field. The experiments compare six global terrain configurations shown in Figure 2: **Ocean, Beach, Island, Lake, Isthmus, and Channel**. To quantify scale effects, we evaluate 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grid sizes. When changing grid size, we scale the initial population in proportion to grid area; after initialization, population size is an outcome of the ecological dynamics rather than a fixed input.

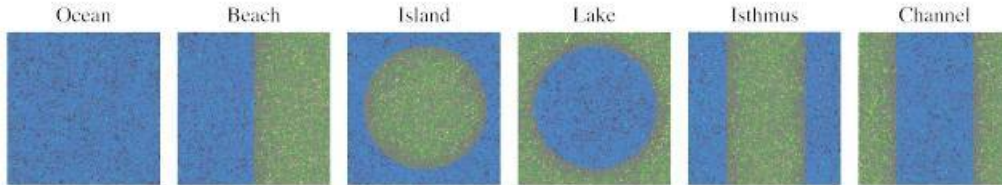


Figure 2: Overhead comparison of the terrain configurations used in the experiments at 256×256 grid size. Dark spots mark individual agents, while green and yellow indicate concentrations of energy and biomass. Each snapshot is taken 5000 steps into an episode, after most water biomass has been eaten but before agents have evolved longer-term land survival.

Resources and Agent Health: Agents must collect **water**, **energy**, and **biomass** to survive and reproduce. At initialization, biomass and energy are randomly distributed across the environment;

during an episode, individual agents consume and redistribute them, making resource location a central measurable outcome of population behavior.

Each action costs a small amount of energy and water. Spent water returns to the landscape and flows downhill to the nearest body of water, whereas spent energy is destroyed. Biomass controls reproduction: every agent requires a fixed amount of biomass to exist, and reproduction succeeds only after the parent has accumulated enough excess biomass to donate to offspring. Total biomass is conserved like water, but unlike water it does not flow or move on its own; energy is not conserved and instead slowly grows on free-standing biomass. These resource dynamics make the empirical task richer than collecting a single food source because agents must balance three measurable resources whose spatial distributions change differently. For ablations on resource rules, see Appendix F.2

Agents also maintain health points (HP), which determine survival and mediate predation. Over-spending a critical resource rapidly reduces HP, while agents can slowly recover HP by using energy and water. Agents may attack other agents, reducing the targets' HP, and they also take a small but increasing amount of age-based damage, so every agent eventually dies of old age if no other cause intervenes. When an agent dies, all resources it carried return to the grid cell where it stood; consequently, a successful attack can convert another agent into consumable resources, yielding a simple predation mechanism. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents observe the environment through four sensor types, and the experiments use these sensors as controlled comparison conditions. The **internal** sensor reports age, health, and carried resources; the **external** sensor reports external resources in the agent's current grid cell; a **compass**, inspired by magnetoreceptors in birds and other animals, reports global direction as a 4-way one-hot vector; and **vision** provides a 7×7 top-down rendered patch around the agent, with three color channels plus a fourth channel for local elevation difference. For ablations on vision configurations, see Appendix F.6

Actions: Agents choose from a finite discrete action set grouped into five categories. **Rest** has no behavioral effect and has the lowest energy cost. The three **Move** actions rotate clockwise, rotate counterclockwise, or move forward one grid cell; because multiple agents cannot occupy the same cell, conflicting moves into occupied or jointly targeted cells are canceled. The **Attack** action kills all agents in the 3×3 square directly in front of the acting agent. For ablations on attack strength, see Appendix F.7. The three **Eat** actions correspond to water, energy, and biomass; when the selected resource is present in the current cell, the action transfers a fixed amount from the environment into the agent's finite-capacity stomach.

The final action, **Reproduce**, creates a new offspring through single-parent reproduction. This action succeeds only when the parent has enough resources to donate to the child. The new agent appears directly behind the parent, immediately receives a fixed amount of the parent's resources, and inherits a mutated copy of the parent's policy. These mechanics make reproduction rate, population stability, and lineage persistence measurable consequences of the same action set.

In the experiments below, we disable selected sensors or the attack action to isolate the ecological effect of different abilities. The three sensing combinations are internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). The suffix **+A** indicates that attacking is enabled; for example, (**RC+A**) denotes agents with resource sensors, a compass, and attacks.

3.3 POLICIES

Agent policies are small, memoryless MLPs, a design that keeps the computational model simple and makes large-population comparisons feasible. Sensor values are flattened, normalized to $[-1, 1]$, and passed through separate linear layers with output dimension 64. The resulting embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer has output dimension equal to the number of available actions, and actions are sampled from a softmax distribution whose temperature evolves with the network weights. We represent weights with half-precision bfloat16 values to support larger populations and faster runtimes. Depending on sensor configuration, each agent policy contains roughly 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5

At the start of each run, agent weights use Kaiming initialization (He et al., 2015) and bias vectors are zero. When an agent is born during an episode, its policy is copied from the parent and then perturbed by Gaussian noise with standard deviation 3×10^{-2} on all mutable weights. This mutation mechanism follows the style of other neuroevolution work (Such et al., 2017). More complex mechanisms such as crossover are possible, but the single-parent mutation rule keeps reproduction simple and makes the scale comparisons easier to interpret. Empirically, the use of low-precision floating-point values required a larger mutation rate; Appendix F.4 reports the mutation-rate ablation.

Agents also carry a 3-channel color trait used when rendering them onto the map. This trait mutates like the network weights and determines how an agent appears to other agents using the vision sensor. In principle, the trait gives visual agents a measurable channel through which they could distinguish different kinds of agents.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and run on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise stated, each seed runs for 2M environment steps or until extinction, so extinction and long-run stability are directly comparable across conditions. Per-seed wall-clock time ranges from under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on one H100 GPU. Appendix E describes simulator architecture, additional scaling measurements, and reproducibility considerations.

4 EXPERIMENTS

The experiments quantify how sensing modality, environmental scale, and population size change emergent behavior. For each comparison, we identify the agent class, terrain, grid size, random-seed count, and outcome metrics so that behavioral differences can be interpreted as changes in consistency, timing, stability, and practical ecological impact.

4.1 LONG-DISTANCE RESOURCE GATHERING

The first experiment measures whether agents evolve long-distance travel inland to collect resources and return to water. We compare compass-equipped agents (**RC**) against the resource-only baseline (**R**) in terrains with clear directional structure. The primary outcome is whether a mining event occurs, and the main empirical pattern is that (**RC**) agents develop more reliable long-range resource gathering than (**R**) agents, with the consistency and effectiveness of the behavior increasing at larger environmental scales.

Figure 3 illustrates the baseline (**R**) dynamics. After 5000 time steps (a), the initial dry-land agents have died from lack of water, while agents in the water have expanded rapidly on the initially rich resources. At this early peak, policies have not yet discovered efficient resource-seeking behavior. Agents then occasionally wander from water onto land (b), but without directional information they often fail to return to the ocean; when they die, their biomass is deposited on land. By 50,000 time steps (c), this repeated transfer has nearly doubled uneaten biomass on land and reduced the water-based resource supply enough to shrink the population. By 125,000 steps (d), almost all biomass has moved to land and the population is near extinction, with the final agent dying around 185,000 steps (e).

This baseline run shows why the quantitative mining metric is ecologically meaningful. The failure of individual agents to return to water changes the spatial distribution of conserved biomass, and that redistribution drives a population-level collapse. The same failure mode also creates a measurable opportunity: if any lineage discovers a policy for walking onto land, eating the accumulated biomass, and returning to water, it can convert the inherited ecological damage into a stable resource-gathering strategy.

Figure 4 shows the matched compass condition (**RC**). The early dynamics are comparable to the baseline: population initially shrinks and free biomass on land rises (a). The divergence appears later, when some agents use the compass to drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” land biomass by navigating between land and water. By

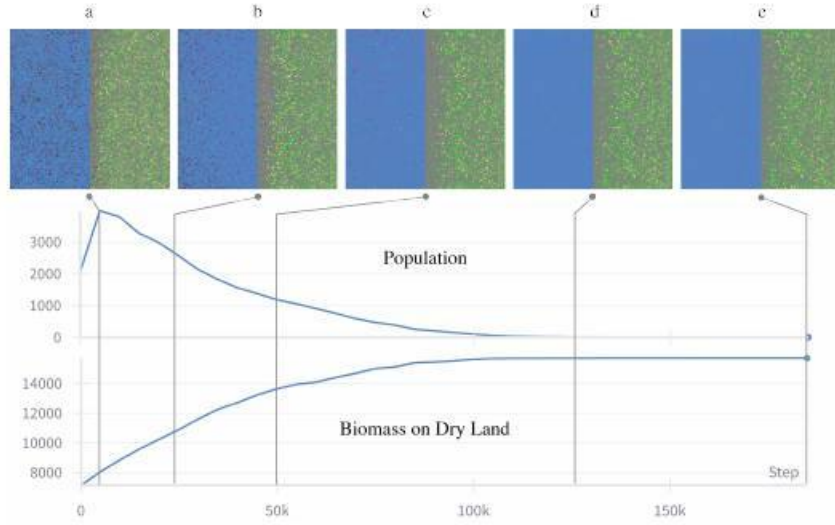


Figure 3: Representative (**R**) baseline trial in a 256×256 **Beach** environment. The top chart tracks population over time, and the bottom chart tracks free biomass on dry land, showing the coupled population decline and land-biomass accumulation when agents lack a compass.

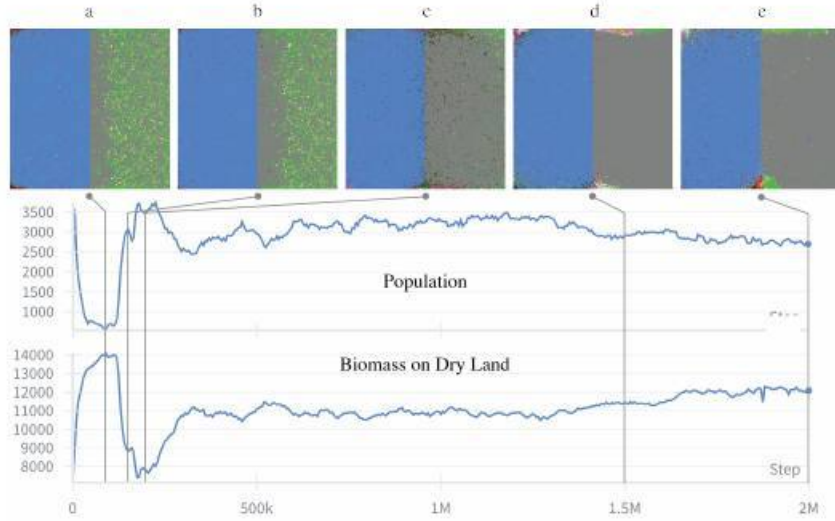


Figure 4: Representative (**RC**) compass trial in a 256×256 **Beach** environment. The same population and dry-land-biomass metrics are shown as in the baseline condition; here, the compass condition supports a mining transition and stable survival through 2,000,000 steps.

200,000 steps (c), these miners have cleared the beach, and although the population later changes shape as agents form edge clumps (d), it remains largely stable through the full 2,000,000 time-step evaluation (e).

We quantify this comparison with **Beach** experiments at five grid sizes: 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , initialized with 32768, 8192, 2048, 512, and 128 agents respectively. Each condition uses four random seeds. We record mining as a binary event when free biomass on dry land drops by at least 10% from its historical maximum, and we also measure mining onset time using the local maximum in land biomass and the resulting percentage drop in biomass.

Table 1: Mining and extinction count comparison for **Beach** worlds across five grid sizes and initial populations. Rows compare compass agents (**RC**) against resource-only agents (**R**);  counts seeds with a mining event and  counts seeds that went extinct before 2M steps. Per-seed onset and drop values are reported in Table 3 in Appendix A

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

Table 1 summarizes mining and extinction counts, with per-seed onset and drop measurements reported in Table 3 in Appendix A. The comparison is sharply separated by sensor condition: (**RC**) agents exhibit mining far more consistently than (**R**) agents. Mining appears in every (**RC**) seed from the 256×256 scale upward, whereas the (**R**) baseline rarely mines and often goes extinct at smaller scales. Thus, the practical effect of the compass is not only earlier discovery of a behavior but also improved population stability, especially as the environment becomes large enough to support the strategy.

To analyze timing rather than only event counts, we run additional (**R**) and (**RC**) seeds at 512×512 world size (32 seeds) and 128×128 world size (64 seeds). We use a Kaplan-Meier estimator to compare the probability of extinction and mining as functions of simulator time. The survival analysis shows that extinction is rarer and delayed at larger world sizes, while mining is more frequent and occurs sooner at smaller grid sizes. A Cox proportional hazards model comparing mining discovery and extinction timing across world size and compass usage gives the same qualitative conclusion: larger worlds substantially reduce extinction rate and enable much more rapid mining discovery than smaller worlds, and adding a compass produces a similar favorable shift. Full details are in Appendix B.

Table 2: Terrain-by-terrain mining and extinction comparison at 512 × 512 grid size. Rows compare compass agents (**RC**) against resource-only agents (**R**) in **Island**, **Lake**, **Isthmus**, and **Channel**;  counts seeds with mining and  counts seeds extinct before 2M steps. Per-seed details are reported in Table 4 in Appendix A

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We next hold grid size fixed at 512 × 512 and compare (**R**) and (**RC**) agents across four additional terrains: **Island**, **Lake**, **Isthmus**, and **Channel**. Table 2 reports mining and extinction counts, and Table 4 in Appendix A gives per-run details. The terrain comparison preserves the main sensor pattern: compass-equipped (**RC**) agents show earlier and more consistent mining, together with lower extinction rates, than (**R**) agents. The margin varies by terrain: (**RC**) agents adapt more reliably on **Island** and **Isthmus** than on **Lake** and **Channel**, where some seeds still go extinct. A likely explanation is that moving straight east or west reliably returns an agent to water in the **Island** and **Isthmus** terrains, whereas **Lake** and **Channel** lack an equally simple directional rule. The (**R**) baseline does not reliably adapt in any terrain, although it struggles least on **Island**. Figure 12 in Appendix D visualizes mining in the island terrain.

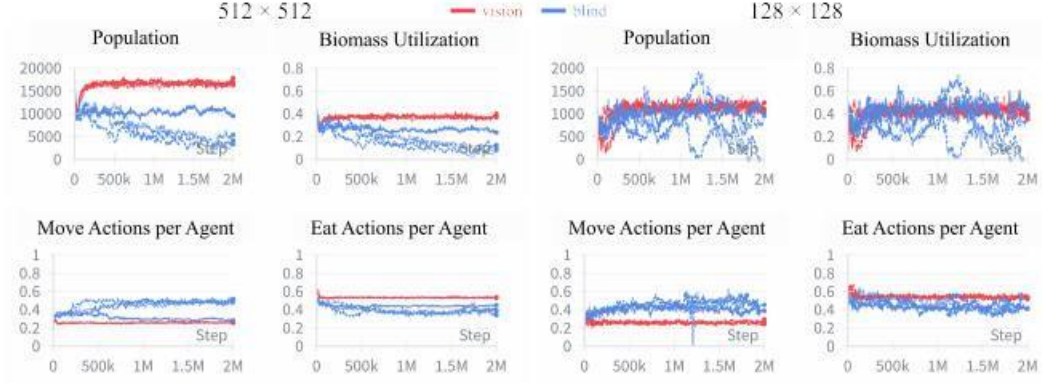


Figure 5: Foraging comparison in 512×512 and 128×128 **Ocean** worlds. Each panel reports one metric—population, biomass utilization, move actions per agent, or eat actions per agent—with biomass utilization and action rates averaged across agents at each time step. Four seeds with vision agents (**RCV + A**) are shown in red and four matched blind-agent seeds (**RC + A**) in blue. The vision condition supports larger, more stable populations, higher biomass utilization, less movement, and more frequent effective eating; the margin is clearest at larger scale, with the 1024×1024 and 256×256 comparison shown in Figure 9 in Appendix C

4.2 VISUAL FORAGING AND PREDATION

The second experiment evaluates whether visual observations improve resource acquisition and predation in the **Ocean** environment. We compare (**RCV+A**) agents, which have resource sensing, a compass, a 7×7 visual sensor, and attacks, against (**RC+A**) agents, which have the same resource, compass, and attack capabilities but no vision. The metrics are population, biomass utilization, movement actions per agent, eat actions per agent, attacks per agent, and homicides per attack.

Figure 5 compares foraging-related metrics at 512×512 and 128×128 grid sizes. Figure 9 in Appendix C provides the corresponding 1024×1024 and 256×256 comparison, and Table 5 in Appendix A reports summary statistics. Biomass utilization is the percentage of total environmental biomass currently held inside agents. Relative to blind (**RC+A**) agents, vision-equipped (**RCV+A**) agents spend less time moving, eat more frequently, support larger populations, and achieve higher biomass utilization. These consistent margins indicate that visual sensing is used for more efficient foraging. The scale pattern is also favorable: run-to-run variance decreases in larger environments, and the vision condition exhibits more stable trajectories with less temporal variation in the measured metrics.

Figure 6 reports predation-related metrics for the same runs. Vision-equipped agents attack relatively infrequently but maintain a successful kill rate of approximately 75%, indicating high precision conditional on attacking. Blind agents in large grids sometimes also reach high kill rates, but those trajectories are much less stable. Visual inspection (see Figure 13 in Appendix D) suggests one confound: some high blind-agent kill rates occur when agents cluster densely in a single location, whereas such clusters almost never form in vision-agent runs. This pattern is consistent with the possibility that vision also helps agents avoid one another, although the causal mechanism remains difficult to isolate. At smaller scales, the difference between visual and blind agents is less pronounced, but visual populations remain far more stable.

4.3 ABLATIONS

Appendix F tests how sensitive the main empirical patterns are to simulator configuration. The ablations separately vary initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). Appendix F.3 further checks whether the large-environment effects can be explained by self-averaging alone; the comparison is designed to distinguish genuine scale-dependent dynamics from averaging over many smaller simulations.

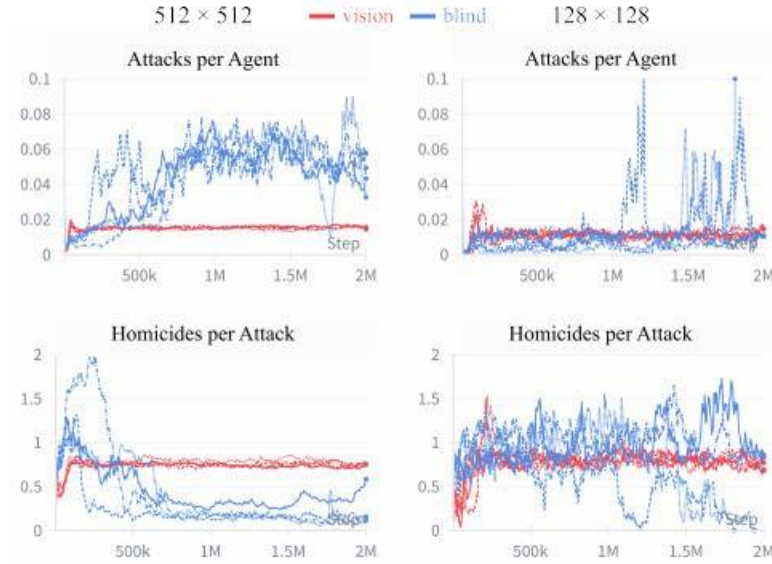


Figure 6: Predation comparison in 512×512 and 128×128 **Ocean** worlds. The metrics are attacks per agent and homicides per attack, averaged across agents. Four seeds with vision agents (**RCV + A**) are shown in red and four blind-agent seeds (**RC + A**) in blue. Vision reduces attack frequency while preserving much higher attack precision, consistent with specialized predation behavior; the separation is stronger at larger scale. Figure 10 in Appendix C shows the corresponding 1024×1024 and 256×256 comparison.

5 CONCLUSION

Our results provide quantitative evidence that open-ended ecological environments are a practical framework for studying the emergence of intelligence. Across controlled comparisons over sensor configurations, terrain types, population sizes, and grid scales, both environmental scale and sensory richness consistently shift the behaviors that evolve, including long-distance resource gathering, vision-based foraging, and predation. Taken together, these empirical patterns show that ecological and evolutionary pressures, when paired with richer environments, can provide a natural route toward increasingly capable artificial agents. The experiments also demonstrate practical significance: large populations with distinct neural policies can evolve rich behaviors within a reasonable hardware budget and without explicit objectives. We hope these results motivate further quantitative study of ecology as both a mechanism for developing artificial intelligence and a controlled lens for understanding it.

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A ADDITIONAL PER-SEED DATA

Table 3: Per-seed mining comparison in the **Beach** world for compass agents (**RC**) and resource-only agents (**R**) across five grid scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			
2	x			n/a	x			1070k	x			165k	x			55k	x			
3	x			n/a	x			280k	x			125k	x			90k	x			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			

Table 4: Per-seed terrain comparison of mining and extinction in **Island**, **Lake**, **Isthmus**, and **Channel**. Compass agents (**RC**) are compared against resource-only agents (**R**) at 512×512 grid size with initial population 8192.

Island					Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Per-seed population and foraging metrics for vision and blind agents in the **Ocean** world across grid scales. denotes population size; denotes biomass utilization, the percentage of total biomass currently inside agents; denotes move actions per agent; and denotes eat actions per agent. Reported values are seed-level averages over all steps, enabling direct comparison of vision versus blind conditions at each scale.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Per-seed predation metrics for vision and blind agents in the **Ocean** world across grid scales.  denotes attacks per agent, averaged across agents and time, and  denotes homicides per attack. Reported values are seed-level averages over all steps. The 0.562  outlier for seed 3 in the 128x128 (RC+A) condition occurs because that seed went extinct before 2M steps, causing attacks per agent to spike as population dwindled.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare extinction and mining-emergence rates as functions of world size and access to global direction through the compass sensor. Extinction is the event that population reaches zero individuals, and mining uses the same threshold as the main experiments: a 10% drop in free biomass on land.

We group runs by both world size and compass access. The dataset contains 64 runs each for 128×128 without a compass and 128×128 with a compass, and 32 runs each for 512×512 without a compass and 512×512 with a compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that did not go extinct within 2M steps as right-censored. Figures 7 and 8 show the resulting comparison: larger worlds and compass access both shift populations toward longer survival and faster mining emergence.

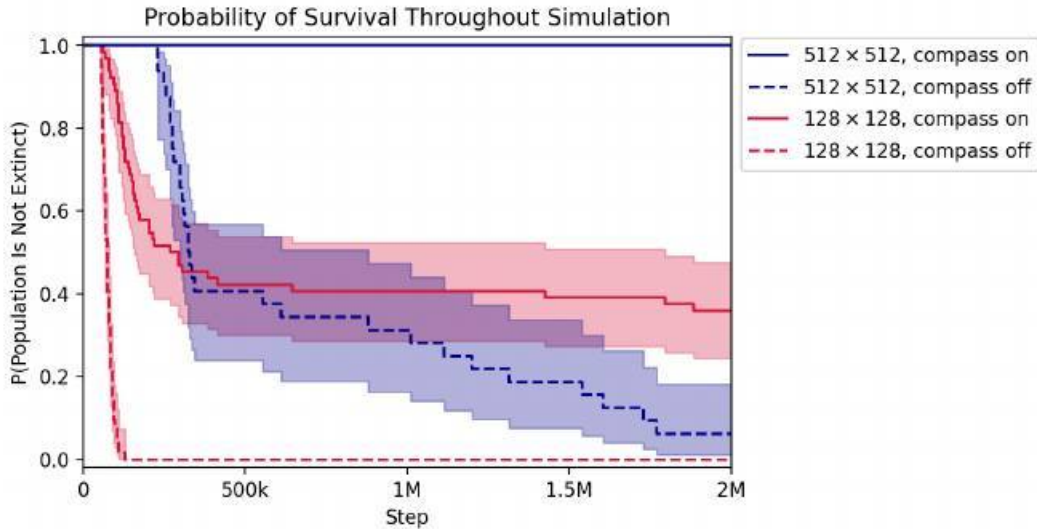


Figure 7: Kaplan-Meier survival curves comparing the probability that the population has not gone extinct as a function of timestep, world size, and compass access. Each group (for example, 512×512 with compass on) has its own non-parametric estimator, with shaded 95% confidence intervals. Each 512×512 group uses 32 runs, and each 128×128 group uses 64 runs.

Table 7 reports Cox proportional hazards models for emergence and extinction timing, treating runs without the relevant event by 2M steps as censored. The hazard ratios make the margins clear. For extinction, larger world size and compass use both substantially reduce hazard: the extinction rate in 128×128 is about 25 times greater than in 512×512 worlds ($HR=0.04$), and enabling the compass gives a similar reduction ($HR=0.03$). For mining emergence, larger worlds accelerate discovery ($HR=6.07$), and compass access produces an even larger acceleration ($HR=34.59$). The models separate groups well, with concordance 0.80 for extinction and 0.85 for mining. These estimates support the interpretation that larger worlds sustain larger populations, lowering the chance of reaching the absorbing zero boundary while also exploring a larger phenotype space through more mutation events than smaller worlds.

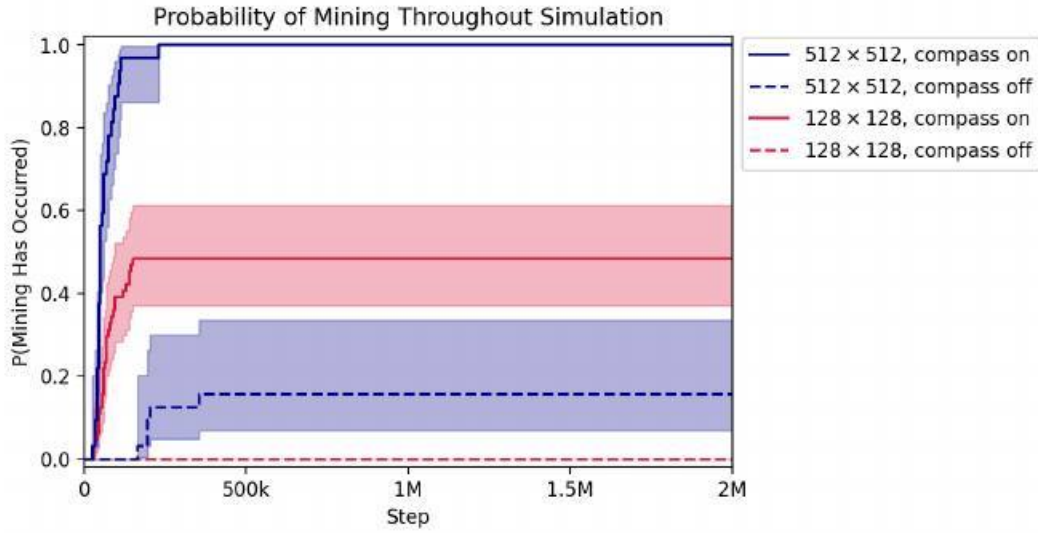


Figure 8: Kaplan-Meier curves comparing the probability that mining has evolved as a function of timestep, world size, and compass access. Each group (for example, 512×512 with compass on) has its own non-parametric estimator, with shaded 95% confidence intervals. Each 512×512 group uses 32 runs, and each 128×128 group uses 64 runs.

Table 7: Cox proportional hazards models for extinction and mining-emergence timing, using the 128×128 compass-off condition as the baseline. Hazard ratios (HR) below 1 indicate reduced hazard and therefore longer time to event; for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the 128×128 baseline rate. The analysis includes 192 total runs: 32 compass-on and 32 compass-off seeds in the 512×512 world, plus 64 compass-on and 64 compass-off seeds in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

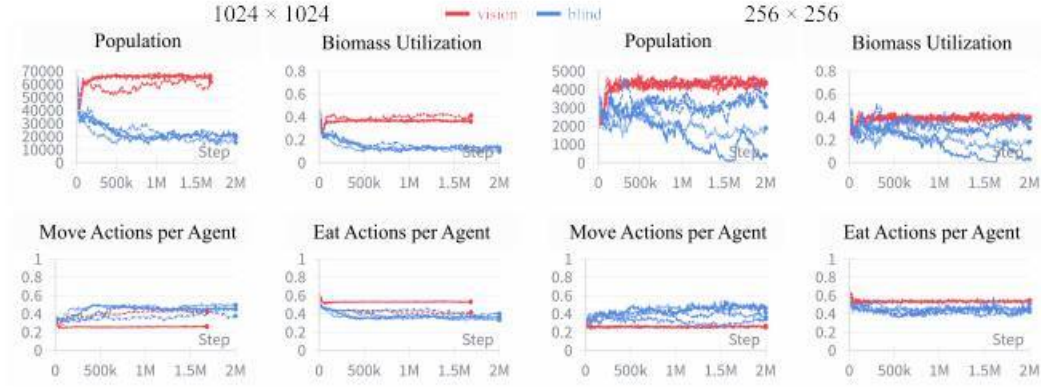


Figure 9: Foraging comparison in the **Ocean** world at 1024×1024 and 256×256 scales. Vision agents (red) support more stable populations and higher biomass utilization than blind agents by moving less and eating more frequently and effectively. The large-scale condition shows lower variance across seeds, reinforcing the main-text pattern that vision and scale jointly improve foraging stability. Due to GPU-hour limits, the 1024×1024 vision seeds ran for 1.7M rather than 2M steps.

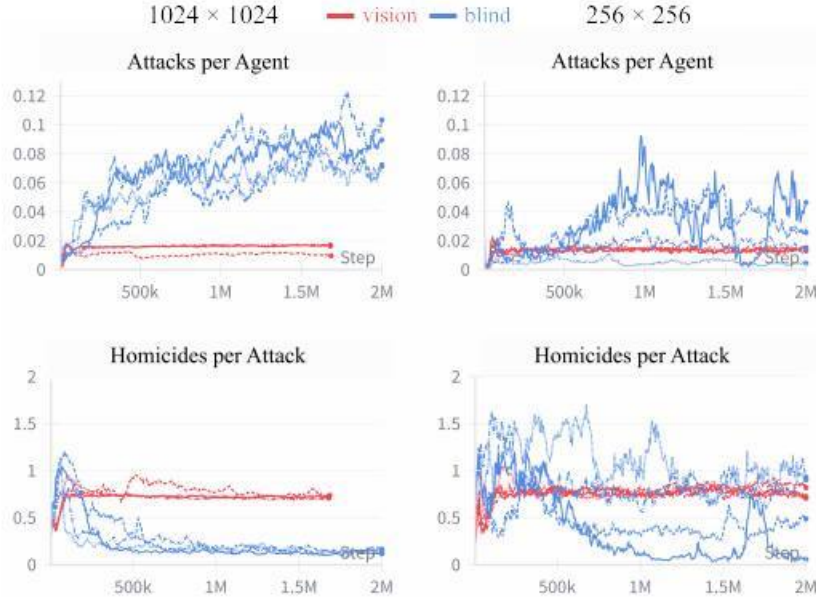


Figure 10: Predation comparison in the **Ocean** world at 1024×1024 and 256×256 scales. Vision agents (red) attack less frequently but with higher homicides per attack, matching the specialized-predation pattern observed alongside improved foraging in Figure 9. The separation is strongest at larger scale, while blind agents (blue) show higher variance at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

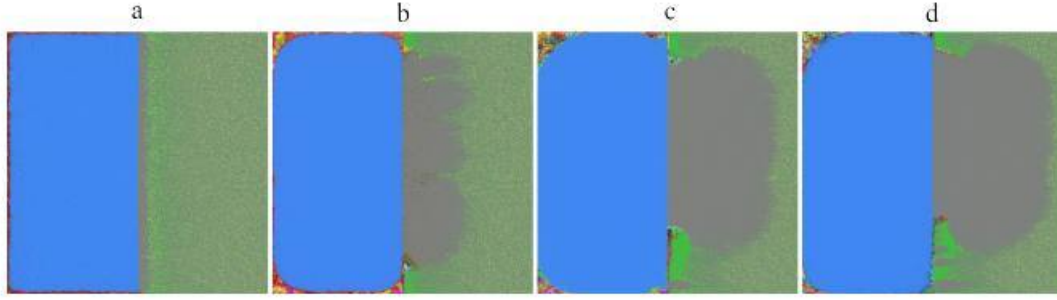


Figure 11: Representative 1024×1024 **Beach** run showing the richer spatial patterns that appear at large scale. (RC) agents first die at the shore (a), transferring biomass to dry land, then develop mining behavior that clears biomass from the beach (b). When biomass later re-accumulates at the beach ends (c), subpopulations again mine these smaller biomass deposits by traveling inland to extract resources (d).

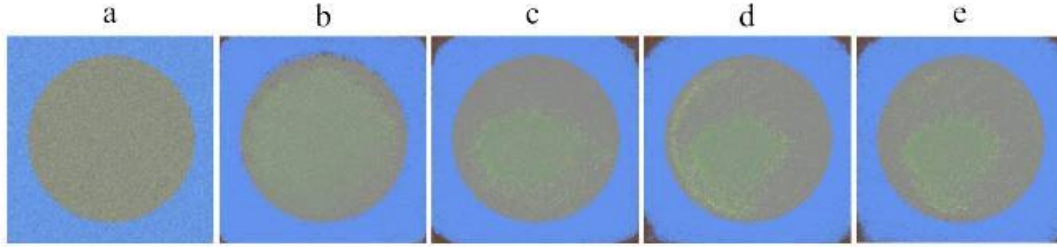


Figure 12: Representative 512×512 **Island** run showing that the mining metric corresponds to repeated spatial resource extraction. With (RC) agents, multiple shoreline regions are mined during the run (b) (c). When agents later die before returning to water, biomass is transferred back onto the island (d), after which agents adapt again and clear the accumulated shore biomass (e).

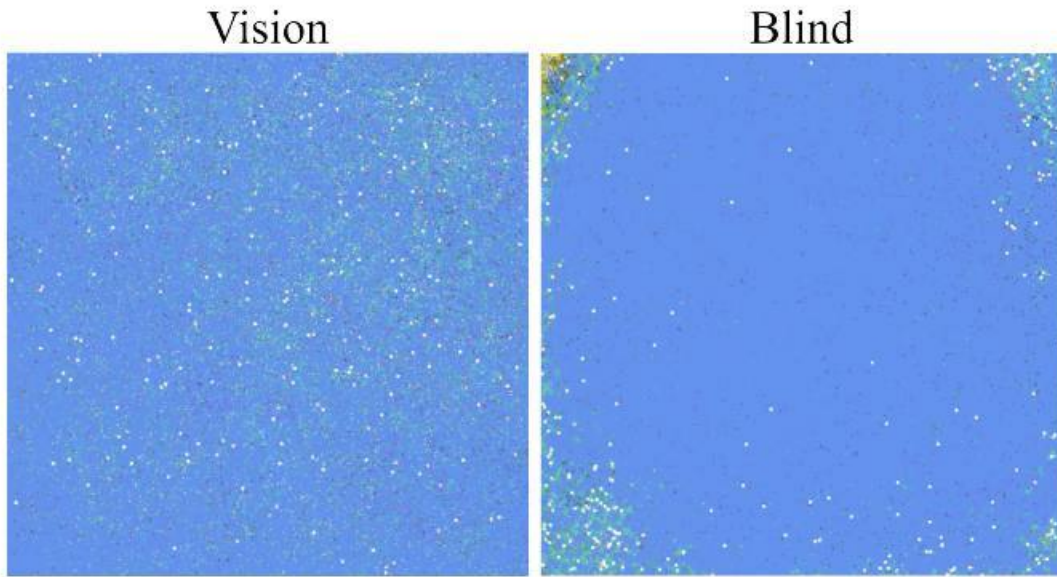


Figure 13: Representative 512×512 **Ocean** comparison with attacks enabled: vision (RCV+A) agents are shown on the left and blind (RC+A) agents on the right. White squares mark attack locations. The visual comparison matches the quantitative results in Figures 5 and 6: vision agents spread through the ocean to locate free resources and attack more effectively, whereas blind agents explore less and more often cluster in world corners.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report the simulation design, runtime measurements, and reproducibility considerations that support the quantitative comparisons in the main paper.

Simulator design. All experiments use a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. The simulator computes each environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so policy evaluation and simulation share one JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation lets the same environment implementation support different evolutionary algorithms, reinforcement learning methods, or other population update rules. Pure JAX functions, vectorization, and just-in-time compilation fuse operations into a small number of GPU kernels, providing the throughput needed for the scale comparisons reported in Table 8.

A key feature of the simulator is built-in measurement tooling. The measurement API records the aggregate statistics used throughout the figures and tables, including biomass utilization, movement and eating rates, attack and homicide rates, and environment-level parameters at each step. An interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools link scalar metrics to qualitative behavior, which was important for validating phenomena such as long-range resource mining and predation patterns.

From a high-performance-computing perspective, the simulator fills a gap relative to existing tools. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), and Griddly (Bamford et al., 2020) are configurable but require substantial environment engineering and do not directly target open-ended, evolving populations at the scales measured here. Conversely, GPU-native multi-agent RL environments typically emphasize fixed agent counts and task-specific rewards. Our simulator combines GPU-native execution, births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination makes large ecological experiments practical and provides a foundation for future quantitative studies of emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 compares runtime across grid sizes and maximum agent populations. Each run uses a single H100 GPU and a maximum horizon of 2M steps. The table reports (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total); runtime can vary with network size, enabled sensors, and logging level.

At the largest 1024×1024 grid size, a **Beach** run takes ~ 10 hours and an **Ocean** run takes ~ 14 hours on one H100, with memory usage of roughly 60GB. The **Ocean** condition is slower because it includes attack actions and vision sensing, making the runtime comparison a practical measure of the added cost of the richer evaluation condition.

Table 8: Runtime comparison for **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each (RC) agent has a 22,465-parameter neural network, and each (RCV + A) agent has a 22,529-parameter neural network. The table reports simulation steps per second (hz) and extrapolated wall-clock hours for 2M steps; effective runtime is lower for seeds that go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Because the environments are large and randomized, precise trajectories can differ substantially across runs, so the reported evidence emphasizes repeated seeds and aggregate patterns rather than single-run exact reproduction. We report four random seeds for each main experiment. Low-level GPU nondeterminism means that repeating a run with the same random seed does not always produce identical trajectories: parallel floating-point operations may execute in different orders or be fused differently, changing rounding behavior. Small finite-precision differences can accumulate over many steps into visible trajectory drift, and 16-bit weights can coarsen these effects without being the primary source. Despite this trajectory-level variability, the same high-level comparisons recur across seeds, which is why the paper foregrounds event frequencies, summary metrics, confidence intervals, and cross-condition margins.

F ABLATIONS

We present ablations that quantify how specific hyperparameters affect the main outcomes: mining frequency, extinction frequency, population size, foraging metrics, and predation metrics under controlled environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We first isolate initial population size in a 256×256 **Beach** world and measure mining consistency and population stability for (**RC**) agents. The corresponding no-compass (**R**) baselines show a clear contrast: they have 0% mining and 100% extinction at every initial population size except 8192, where they reach 25% mining and 75% extinction. Mining and extinction frequencies are not complementary because a run can mine and later go extinct, or survive the full 2M steps without discovering mining.

Figure 14 shows that initial population size has little measurable effect on either mining consistency or extinction rate in this setting. Populations initialized with only 128 agents mine as frequently as those initialized with 8192 agents, and extinction rates remain largely stable across the sweep.

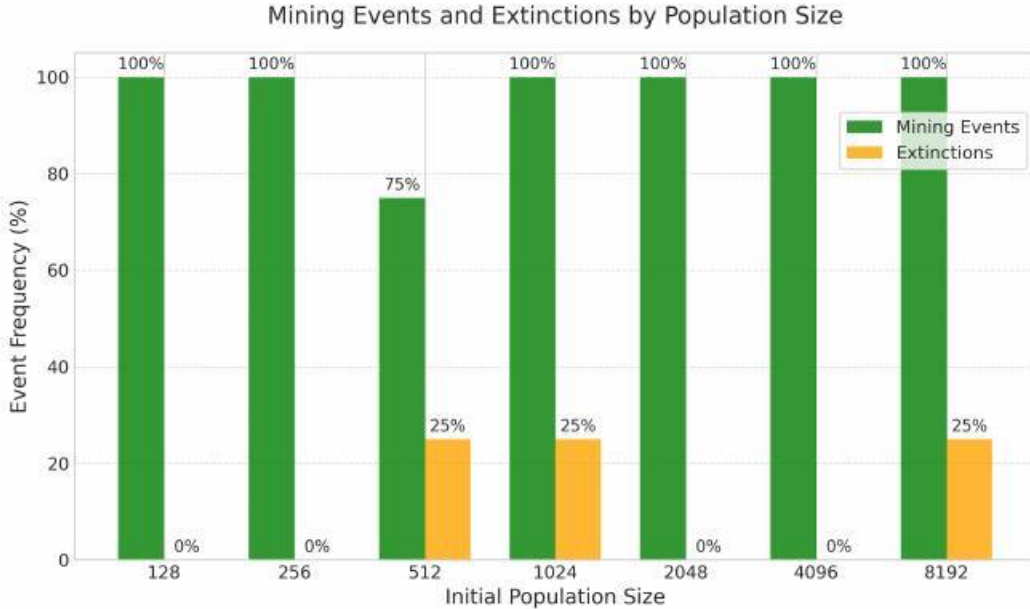


Figure 14: Mining and extinction frequencies in a 256×256 **Beach** world as initial (**RC**) population size varies. Each frequency is computed over four random seeds; for example, 25% extinction means that 1/4 seeds reached population 0 before 2M simulation steps.

Figure 15 explains why the event frequencies remain stable across initial population sizes. Regardless of initialization, each run rapidly approaches a similar population size within the first 50,000

simulation steps, indicating an implicit carrying capacity determined by the fixed resources in the environment. Populations initialized below this capacity grow toward it, while populations initialized above it are constrained downward, so the effective population size is similar after the early transient. Larger initial populations still add some resources because agents carry biomass that is deposited upon death, and therefore reach somewhat higher maximum populations on average over the full simulation (Table 9); however, the margin is not large enough to change mining or extinction frequency in this setting. Later in a run, populations can diverge substantially as different evolved abilities alter resource distributions at different times.

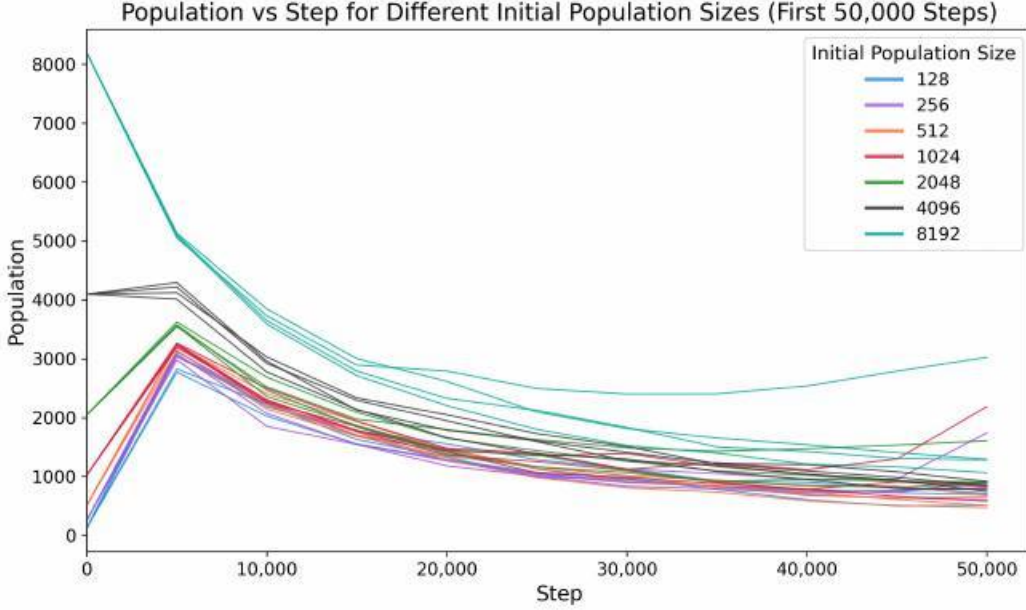


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with **(RC)** agents, with four seeds per initial population size. All initial population sizes approach a similar early carrying capacity determined by available resources. Larger initial populations contribute additional biomass through agent deaths, raising the maximum supported population slightly.

Table 9: Maximum population size after step 0 for each starting population size, reported as mean and standard deviation over four random seeds. Larger initial populations reach slightly higher maxima and show greater seed variance, but the maximum is mostly robust: the 128-agent initialization, only 1.6% of the 8192-agent initialization, still reaches 78.9% of the latter maximum population (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

F.2 VARYING RESOURCE RULES

We next vary two resource hyperparameters, biomass photosynthesis rate and initial biomass site density, and measure their effects on mining and extinction frequency. The setting matches Appendix F.1 with initial population fixed at 2048 and compass agents **(RC)**.

The biomass photosynthesis rate controls how quickly landscape biomass creates energy over time, while initial biomass site density controls the fraction of grid cells containing biomass at simulation start. The main experiments in Section 4 use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Unlike initial population size (Figure 14), the resource-rule sweep produces clear differences in mining and extinction consistency, as shown in Figures 16 and 17.

The quantitative pattern follows the carrying-capacity interpretation: higher biomass photosynthesis rates or initial biomass site densities increase available resources, which supports larger populations, increases opportunities for mining-favorable mutations, and lowers extinction risk. The population margins support this mechanism. Omitting extinct seeds, runs with initial biomass site density 1/16 reached an average maximum population of 1608 after step 0, compared with 2466 at density 1/8, 3786 at density 1/4, and 5353 at density 1/2.

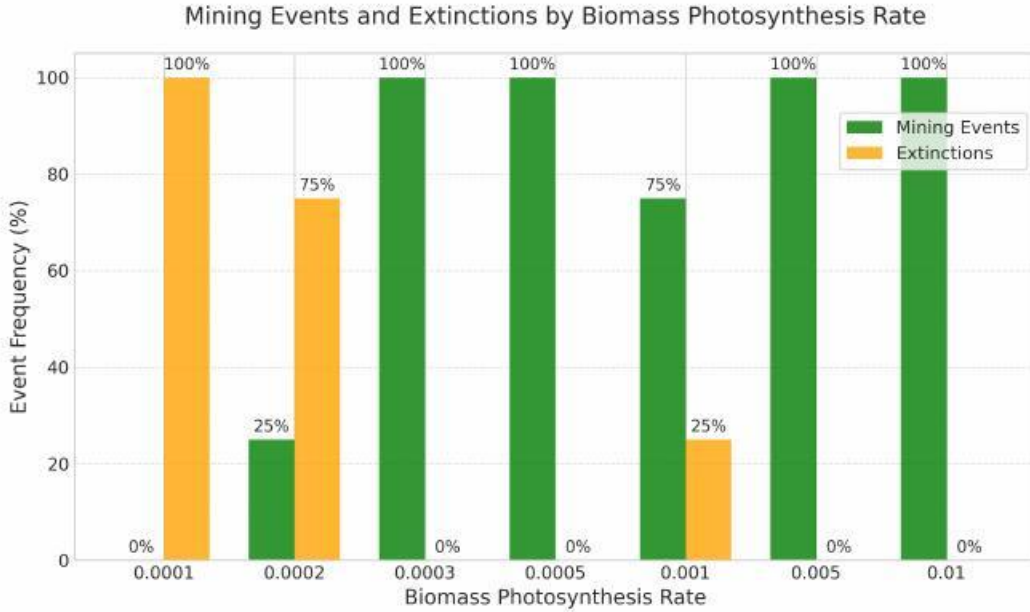


Figure 16: Mining and extinction frequencies in a 256×256 **Beach** world as biomass photosynthesis rate varies. Each frequency is computed over four random seeds. Mining consistently emerges once the biomass photosynthesis rate reaches 0.0003 or greater.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One alternative explanation for the improved mining stability in larger environments (Table 1) is self-averaging rather than qualitatively different dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so a single large system behaves like the average of many smaller systems of the same total size. Under that view, a 512×512 **Beach** world would be equivalent to many loosely coupled 128×128 worlds in parallel, and the observed scaling results could be reproduced by averaging enough small-world seeds.

To test this explanation, we hold the total “sample size” roughly fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we compare three allocations:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

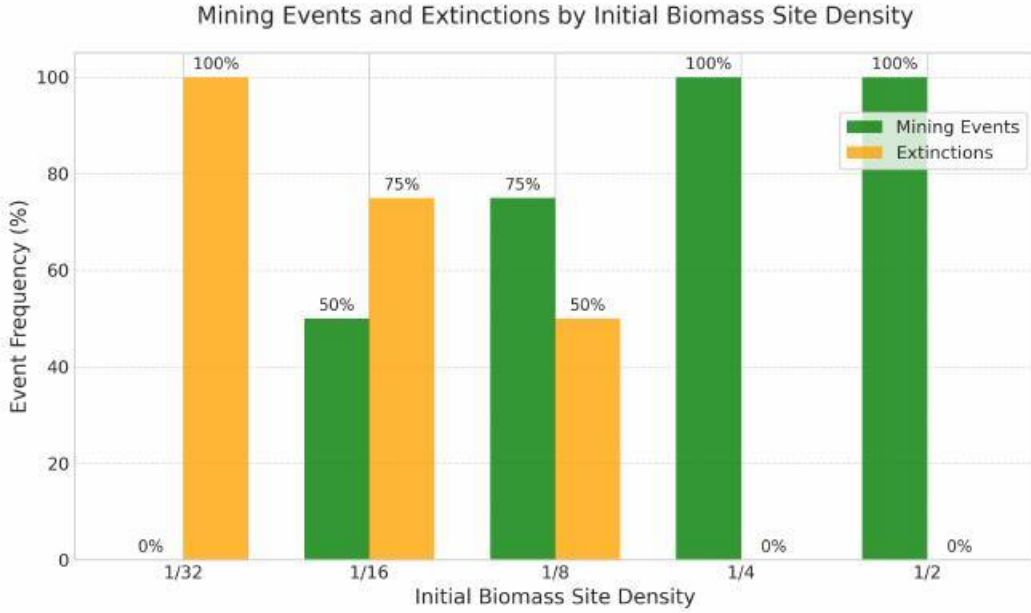


Figure 17: Mining and extinction frequencies in a 256×256 **Beach** world as initial biomass site density varies. Each frequency is computed over four random seeds. Higher initial biomass density yields more consistent mining discovery and lower extinction risk.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed runs for 2M steps with the same hyperparameters as Section 4.1. As a control, we repeat the protocol for no-compass agents (**R**). Table 10 reports long-distance mining and extinction frequencies for each allocation.

If large environments were merely self-averages of smaller ones, mining and extinction rates should be approximately invariant to the allocation: 4 large worlds and 64 small worlds would have similar mining probabilities after matching total cells and agents. Instead, the event counts depend strongly on world size. For (**RC**) agents, mining appears in 4/4 runs with 0/4 extinctions at 512×512 , remains frequent but imperfect at 256×256 with 15/16 mining and 1/16 extinction, and becomes fragile at 128×128 with 31/64 mining and 41/64 extinction. The (**R**) control never mines and goes extinct at all scales, reinforcing that the compass is a key enabler of long-range behavior.

Table 10: Mining and extinction frequencies when trading off world size against the number of independent runs. We compare **Beach** worlds at 512×512 , 256×256 , and 128×128 with proportionally scaled initial populations of 8192, 2048, and 512 agents. The allocation matches total grid cells and initial agents across conditions by using 4, 16, and 64 seeds respectively. reports mining frequency and reports extinction before 2M steps for compass agents (**RC**) and resource-only agents (**R**).

	512 × 512 8192		256 × 256 2048		128 × 128 512	
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These counts are inconsistent with a pure self-averaging explanation: larger environments enable and stabilize mining in ways that cannot be reproduced by averaging more small simulations.

Even with total cells, initial agents, and simulation steps matched, concentrating resources into a few large worlds yields consistent mining and rare extinction, whereas distributing the same budget across many small worlds yields frequent mining failures and population collapse. This suggests that environmental scale changes the underlying ecological dynamics rather than simply smoothing stochastic noise, plausibly by supporting larger, more spatially distributed populations and more persistent policy diversity.

F.4 MUTATION RATE SWEEP

We test mining sensitivity to the mutation rate, which controls the scale of Gaussian noise added to a child policy at birth and therefore the expected policy distance between parent and child. The evaluation again uses a 256×256 **Beach** world with 2048 initial (**RC**) agents and a maximum horizon of 2M steps.

Figure 18 shows a broad robust interval: for mutation rates from 6×10^{-3} through 7.875×10^{-2} , mining emerges in at least 3 of 4 seeds. This includes the main-experiment value 3×10^{-2} from Section 4.1. Very small mutation rates, such as 3×10^{-3} , limit behavioral innovation and impair survival, while very large mutation rates, such as 3×10^{-1} , destabilize the population and consistently produce extinction without mining.

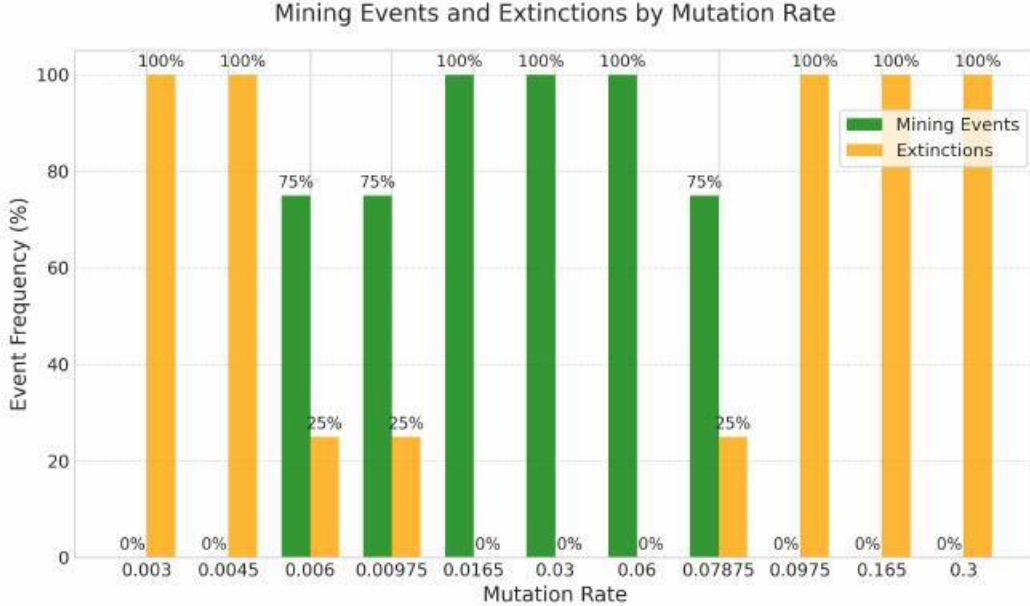


Figure 18: Mining and extinction frequencies in a 256×256 **Beach** world as mutation rate varies. Frequencies are computed over four random seeds per setting, highlighting the interval in which mining remains consistent and the extremes where extinction dominates.

F.5 NETWORK SIZE

We measure how policy-network size and architecture affect mining and extinction, and whether environmental scale changes that sensitivity. The comparison uses 256×256 and 512×512 **Beach** worlds with 2048 and 8192 initial (**RC**) agents respectively, preserving the four-fold area scaling. Matching (**R**) baselines again show rare to no mining and 100% extinction, so the network-size analysis focuses on compass agents.

The main experiments in Section 4 use 2-layer, 64-channel policies with 22465 parameters per agent (Table 11). In the 256×256 **Beach** condition, Figure 19 shows that changing width alone has little

effect: 32 channels (9185 parameters) and 128 channels (61313 parameters) produce similar mining and extinction frequencies when depth is fixed. Depth changes matter more at this scale: all 1-layer, 32-channel populations go extinct, and 50% of 3-layer, 128-channel populations go extinct.

Scaling the environment to 512×512 changes the pattern substantially (Figure 20). In the larger world, 3-layer, 128-channel populations consistently mine and survive through 2M steps, and even the small 1-layer, 32-channel populations mine consistently. We hypothesize that small and large networks face different search difficulties: reduced capacity can make good policies harder to represent, while larger parameter spaces can make useful mutations harder to find quickly. The key empirical result is that architectures that fail often in the smaller environment succeed in the larger one, strengthening the evidence that scale stabilizes behavior. Larger environments support larger populations and more policy-space exploration, while also slowing the spread of locally successful mutants, which can preserve diversity; for related discussion see (Mitchell et al., 2006).

Table 11: Parameter counts for (RC) and (R) agent policies under each network architecture used in the network-size sweep.

Architecture (layers, channels)	Parameter Count (RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

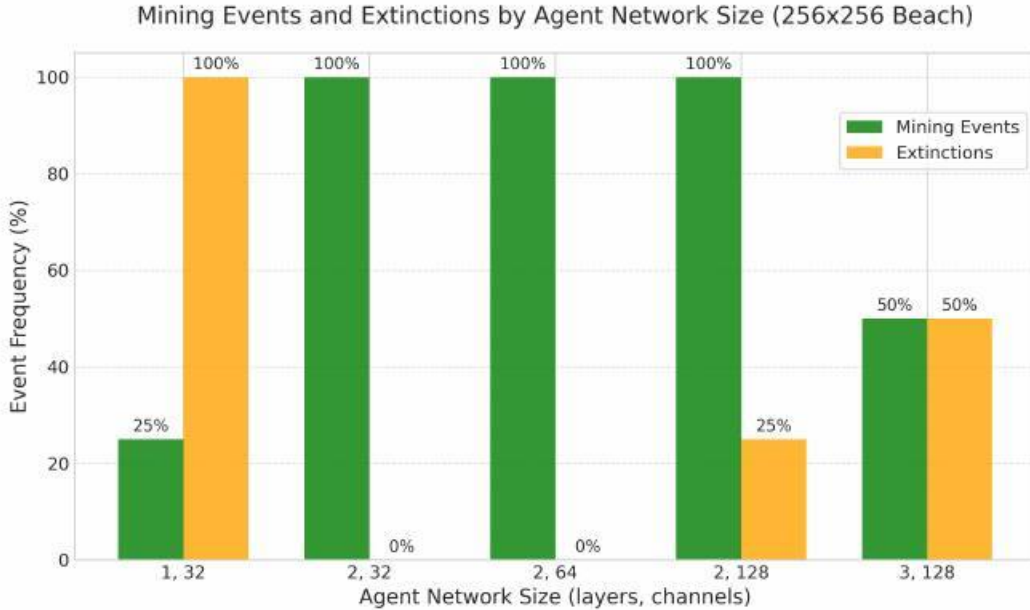


Figure 19: Mining and extinction frequencies in a 256×256 **Beach** world across agent policy network sizes. Each setting uses four random seeds, showing that depth changes are less stable at this smaller scale than width-only changes.

F.6 VARYING VISION RANGE

We quantify how visual range and field-of-view geometry affect foraging and predation in the 512×512 **Ocean** world with (RCV+A) agents. The sweep compares 3×3 , 7×7 , and 9×9 visual ranges. For each range, we evaluate the two configurations in Figure 21 Around, where agents

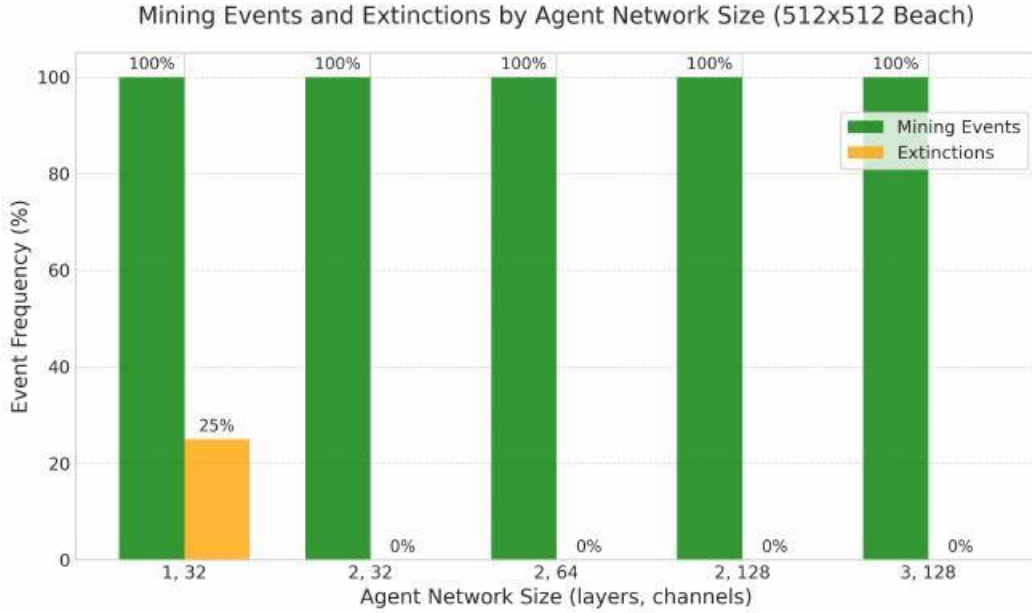


Figure 20: Mining and extinction frequencies in a 512×512 **Beach** world across agent policy network sizes. Each setting uses four random seeds, showing greater robustness to both small and large architectures at the larger scale.

see surrounding cells, and Front, where agents do not see behind themselves. The total number of visible cells is matched between configurations. The main vision experiments in Section 4.2 use the 7×7 Around setting.

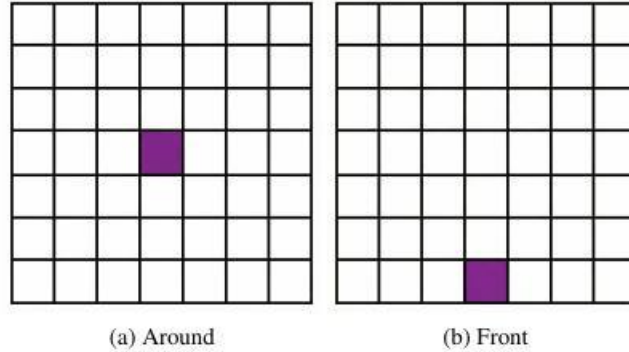


Figure 21: Around and Front viewing configurations for the 7×7 vision range. The purple cell marks an agent facing upward, making the matched visible-cell comparison explicit.

The vision-range comparison shows a threshold followed by diminishing returns. A 3×3 range produces behavior close to the Blind baseline, whereas 9×9 closely matches 7×7 . In this environment, the additional peripheral pixels in a 9×9 memoryless observation are unlikely to change the optimal immediate action often enough to provide a strong learning signal beyond the 7×7 range.

The field-of-view comparison further suggests that Around is slightly more favorable than Front at 7×7 and 9×9 , with higher average population , biomass utilization , and homicides per attack . This margin is consistent with nearby cells carrying more near-term action information than distant forward cells. At 3×3 , however, the direction reverses: Front performs better than Around, suggesting that one additional forward pixel is more valuable than one rear pixel when the overall visual range is extremely small.

Table 12: Foraging and predation metrics in the 512×512 **Ocean** world for (RCV+A) agents under different vision ranges and viewing configurations. Metric icons follow Tables 5 and 6. For each seed, metrics are averaged from step 500k onward to focus on stabilized behavior; reported values are the mean and standard deviation across four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	.181 ± .060	.435 ± .090	.392 ± .033	.032 ± .006	.307 ± .090
	Around	10308 ± 4016	.258 ± .105	.352 ± .014	.427 ± .022	.040 ± .030	.388 ± .195
	Front	13863 ± 520	.351 ± .023	.336 ± .008	.437 ± .009	.026 ± .006	.470 ± .145
7×7	Around	16739 ± 122	.373 ± .003	.259 ± .000	.532 ± .001	.015 ± .000	.751 ± .007
	Front	15255 ± 623	.343 ± .015	.253 ± .001	.540 ± .004	.015 ± .000	.690 ± .041
9×9	Around	16744 ± 117	.372 ± .002	.257 ± .001	.534 ± .001	.015 ± .000	.728 ± .009
	Front	15211 ± 1416	.342 ± .033	.253 ± .001	.541 ± .007	.015 ± .000	.664 ± .050

F.7 VARYING ATTACK STRENGTH

We measure how reducing attack strength changes emergent behavior and population dynamics in the 256×256 **Ocean** world. The sweep compares three settings: Full Attack, which deals 10 HP and instantly kills any agent in the attack radius; Half Attack, which deals 5 HP; and Quarter Attack, which deals 2.5 HP. The main **Ocean** experiments in Section 4.2 use the Full Attack setting.

Reducing attack strength from Full Attack to Half Attack substantially reduces both attack frequency and attack effectiveness: attacks per agent and homicides per attack both drop in Figure 22. Lower attack strength therefore supports larger populations, more movement, more eating, and higher biomass utilization. Moving from Half Attack to Quarter Attack yields smaller additional gains, indicating diminishing returns. The mechanism is a double reduction in violence: weaker attacks require more hits to kill, and because each attack is less useful, attacking behavior is selected for less strongly, so agents become less specialized at predation and spend more time foraging.

F.8 VARYING AGENT HP

We investigate whether reducing child starting HP and lowering the healing rate changes behavior by pressuring agents to “grow” over their lifetime. The healing-rate parameter determines how much HP an agent regenerates per healing step by converting available energy to HP, so agents influence healing indirectly by maintaining enough energy.

We compare two settings. In Full HP, child HP starts at 5 and the healing rate is 1 HP per 0.1 energy. In Low HP, child HP starts at 1 and the healing rate is 0.5. Maximum agent HP is 10 in both settings, and the main experiments in Section 4 use Full HP.

Across the two HP settings, Figure 23 shows no significant difference in homicides per attack, eat-action frequency, or biomass utilization. The visible margins are smaller: Full HP populations grow slightly larger, while Low HP agents attack less often and move slightly more. One plausible explanation is that low-HP agents must prioritize early-life foraging to reach full size, shifting action allocation away from attacks.

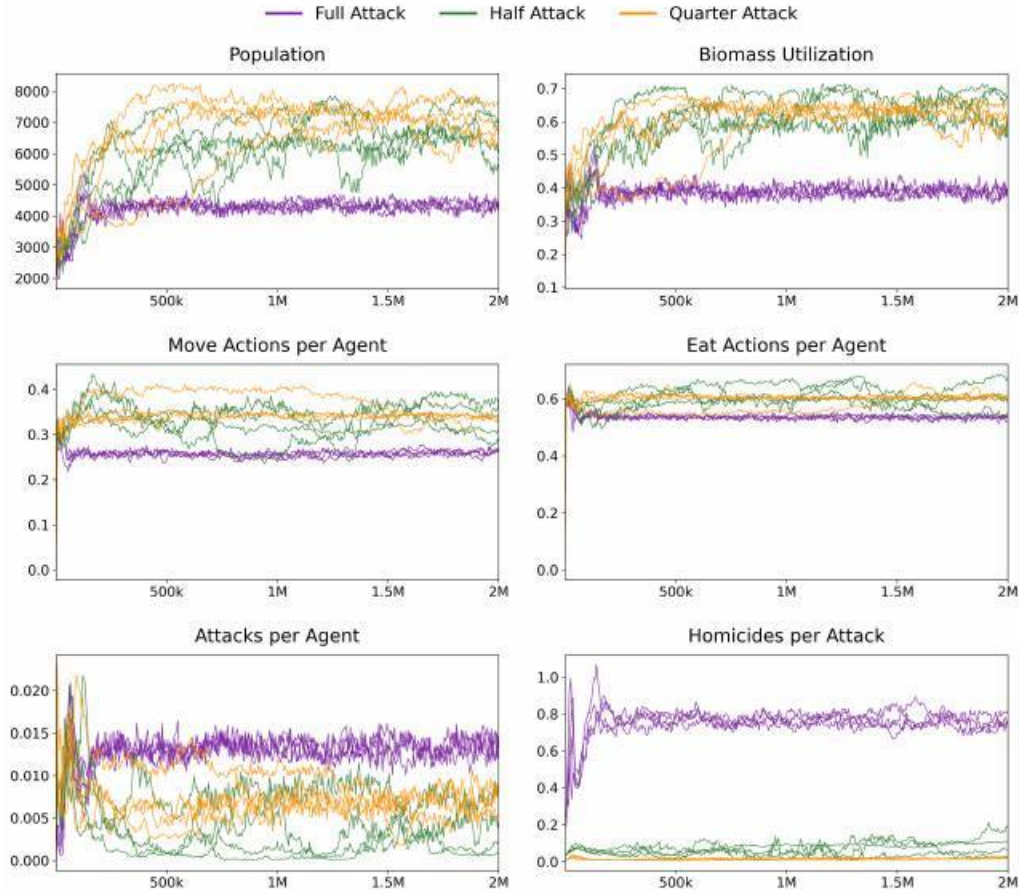


Figure 22: Population and action-frequency comparison over 2M steps in a 256×256 Ocean world for (RCV+A) agents under Full Attack, Half Attack, and Quarter Attack settings. Four seeds are plotted per setting, showing how weaker attacks shift populations toward larger size and more foraging.

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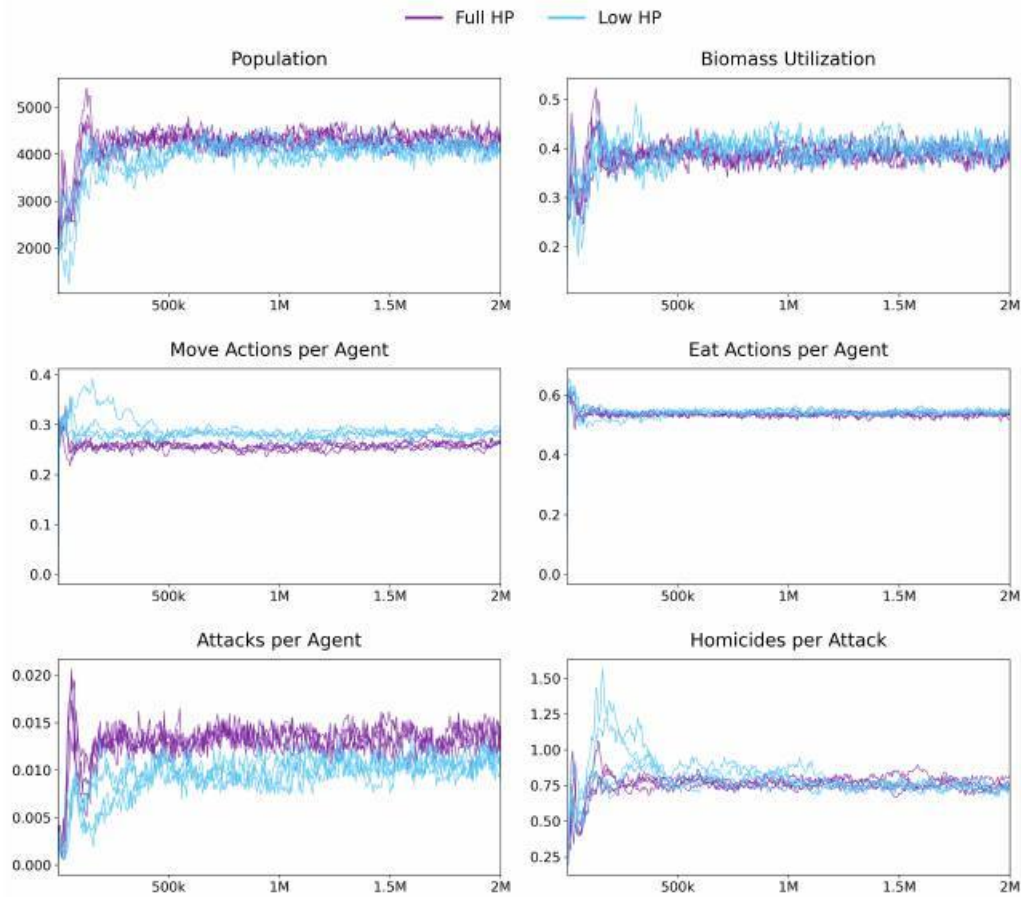


Figure 23: Population and action-frequency comparison over 2M steps in the 256x256 Ocean world for (RCV+A) agents under Full HP and Low HP settings. Four seeds are plotted per setting, showing small action-allocation shifts with little change in foraging or predation effectiveness metrics.